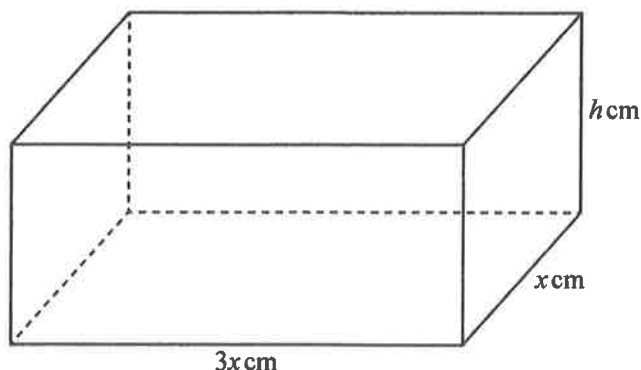


Basic

1. Find the coordinates of the stationary points of the curve $y = x^2 + \frac{4}{x^2}$ and determine the nature of each stationary point. [6]
[N20/I/6]
2. An ice cube of side x cm is melting in such a way that the total surface area, A cm², is decreasing at a constant rate of 48 cm²/s. Assuming that the cube retains its shape, calculate the rate of change of x when $x = 10$. [4]
[N19/I/4]
3. The equation of a curve is $y = x + \frac{2x + 5}{x - 2}$.
(i) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. [3]
(ii) Find the x -coordinate of each of the stationary points of the curve. [3]
(iii) Find the nature of each stationary point. [2]
[N19/I/8]
4. The equation of a curve is $y = 8 - (2x + 1)^3$.
(i) Explain why the curve has only one stationary point and why this is a point of inflexion. [5]
(ii) Write down the coordinates of the stationary point. [1]
[N17/II/8(b)]
5. The equation of a curve is $y = 2 - x^2 - \frac{16}{x^2}$.
(i) Find the coordinates of the stationary points of the curve. [5]
(ii) Use the second derivative test to determine the nature of each of these points. [3]
[N16/I/9]
6. The equation of a curve is $y = \frac{2x^2}{x - 1}$.
(i) Find an expression for $\frac{dy}{dx}$ and obtain the coordinates of the stationary points of the curve. [5]
(ii) Find an expression $\frac{d^2y}{dx^2}$ and hence determine the nature of these stationary points. [4]
[N15/II/6]

7.

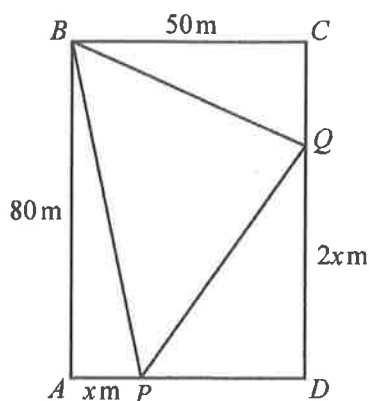


The diagram shows a closed food container designed in the shape of a cuboid. The rectangular base has dimensions of x cm and $3x$ cm. The height of the container is h cm. The volume of the container is 2000 cm^3 .

- Show that the total surface area of the six faces of the container is $6x^2 + \frac{16\,000}{3x}$. [4]
- Given that x can vary, find the value of x for which the total surface area is stationary. [4]
- Determine whether this value of x gives a maximum or minimum value for the total surface area of the container. [2]

[N22/II/4]

8.



The diagram shows a rectangular field measuring 50 m by 80 m. Points P and Q lie on AD and CD respectively. Paths BP , PQ and QB cross the field. The triangular plot BPQ is to be planted with maize. It is given that $AP = x$ m and $DQ = 2x$ m.

- Show that the area, $A \text{ m}^2$, to be planted with maize is given by $A = x^2 - 40x + 2000$. [3]
- Given that x can vary, find the stationary value of A and determine its nature. [4]

[N21/I/8]

- A spherical balloon is being inflated with helium gas at a constant rate of 500 cm^3 per second. The balloon is initially empty.
 - Find the radius of the balloon after one minute.
[The volume of a sphere of radius r is $\frac{4}{3}\pi r^3$.] [2]
 - Find the rate at which the radius of the balloon is increasing after one minute. [3]
 - When a weather balloon is subject to different pressures, the pressure, P atmospheres, and the volume, V litres, of the air in the balloon are related by the formula $PV = k$, where k is a constant. Given that when $P = 1.2$, $V = 2$, find the rate at which V is changing with respect to P when $V = 2$. [4]

[N21/I/13]

10. Water is poured, at a constant rate of $18\pi \text{ cm}^3/\text{s}$, into a hemispherical bowl of radius 12 cm. When the depth of water directly below the centre is x cm, the volume, $V \text{ cm}^3$, of water in the bowl is given by

$$V = \frac{1}{3} \pi x^2 (36 - x).$$

Find

- (i) the time taken for the depth of water directly below the centre to reach 9 cm, [3]
 (ii) the rate of change of the depth of water directly below the centre at this time. [4]

[N18/I/7]

11. A gardener uses 20 m of fencing with which to enclose 2 flower beds. One bed is to be an equilateral triangle of side x m. The other bed is to be a circle of radius r m.

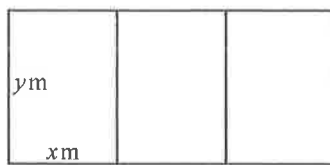
- (i) Express r in terms of x . [1]
 (ii) Show that the total area, $A \text{ m}^2$, of the two flower beds is given by

$$A = \frac{\sqrt{3} \pi x^2 + (20 - 3x)^2}{4\pi}. \quad [3]$$

- (iii) Given that x can vary, find the value of x which gives a stationary value of A . [4]
 (iv) Find the nature of this stationary value and explain why the gardener might be disappointed. [2]

[N18/I/10]

12. A tennis club makes three equally sized tennis courts, positioned next to each other as shown in the diagram below. Each tennis court is rectangular and has sides of length x m and y m. The lines in the diagram represent wire netting. The total length of wire netting used is 288 m.



- (i) Show that the total area, $A \text{ m}^2$, of the three tennis courts is given by $A = 216x - \frac{9}{2}x^2$. [3]
 (ii) Given that x can vary, find the dimensions of each tennis court that make A a maximum. [3]
 (You are not required to show that A is a maximum.)

[N17/I/6]

13. At a given instant, a cyclist is at a cross roads O , travelling due west at a constant speed of 5 m/s. At the same instant a second cyclist is 100 m from O , travelling due north towards O at a constant speed of 10 m/s. This situation is shown in Fig. 1.

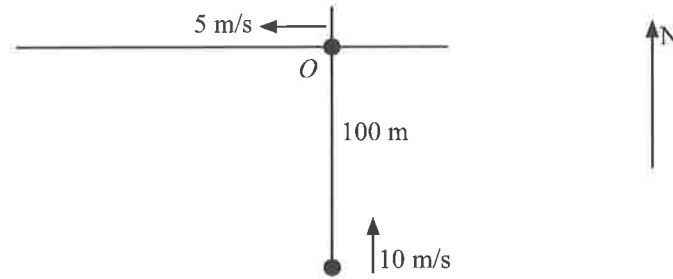


Fig. 1

The position, t seconds later, when the cyclists have reached points P and Q , is shown in Fig. 2.

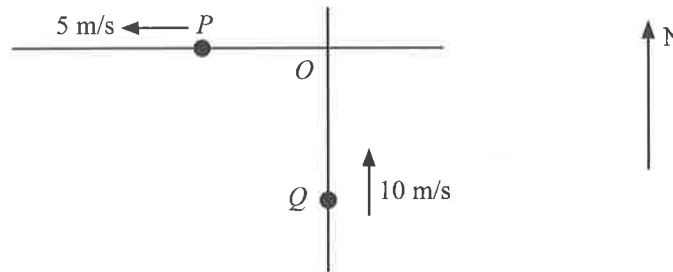


Fig. 2

- (i) Express OP and OQ in terms of t and hence show that the distance, s m, between the two cyclists at time t is given by $s = \sqrt{125(t^2 - 16t + 80)}$. [3]
- (ii) Obtain an expression for $\frac{ds}{dt}$. [2]
- (iii) Find the least distance between the two cyclists. [2]

[N16/I/7]

14. It is given that $y = (x - 2)(2x - 5)^3$.

- (i) Obtain an expression for $\frac{dy}{dx}$ in the form $(ax + b)(2x - 5)^2$, where a and b are integers. [3]
- (ii) Determine the values of x for which y is a decreasing function. [1]

The variables x and y are such that, when $x = 3$, y is increasing at a rate of 0.35 units per second.

- (iii) Find the rate of change of x when $x = 3$. [2]
- It is given further that the variable z is such that $z = y^2$.
- (iv) Show that, when $x = 3$, z is increasing at twice the rate of y . [2]

[N16/II/6]

15. It is given that $f(x) = x^3 - 3x^2 + 4x - 12$.

- (i) By showing clearly your working factorise $f(x)$. [3]
- (ii) Explain why the equation $f(x) = 0$ has only one real root and state its value. [2]
- (iii) Find the value of the constant k for which the graph of $y = f(x) + kx$ has a stationary point at which $\frac{d^2y}{dx^2} = 0$. [5]

[N16/II/8]

16. The equation of a curve is $y = -x^2 + 4x - 6$. The point P lies on the curve and has an x -coordinate of 1. The tangent to the curve at P meets the x -axis at A and the y -axis at B .

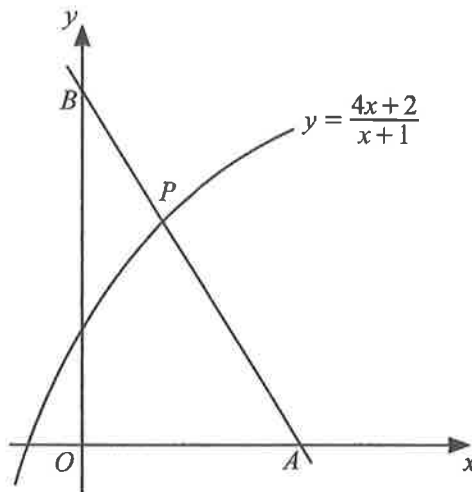
- (i) Find the area of triangle AOB , where O is the origin. [6]

The point Q also lies on the curve. The normal to the curve at Q is parallel to the tangent to the curve at P .

- (ii) Find the coordinates of Q . [3]

[N15/II/3]

17.



The diagram shows part of the curve $y = \frac{4x+2}{x+1}$ for $x > -1$.

- (a) Explain why the curve does not have a stationary point. [3]
- (b) The point P lies on the curve and the gradient of the curve at P is $\frac{1}{2}$. The normal to the curve at P meets the x -axis at A and the y -axis at B . Find the area of triangle AOB . [5]
- (c) The line $y = c$ does not intersect the curve. By expressing $\frac{4x+2}{x+1}$ in the form $a + \frac{b}{x+1}$, where a , b and c are constants, explain why $c \geq 4$. [2]

[N22/I/13]

18. The curve $y = \frac{6}{\sqrt{x}} + x$ has a minimum point M .

- (i) Show that the x -coordinate of M satisfies the equation $x^3 = 9$. [4]
- Points $A(1, 7)$ and $B(4, 7)$ lie on the curve $y = \frac{6}{\sqrt{x}} + x$. The tangents to the curve at A and B meet at the point P .
- (ii) Determine, with working, whether the x -coordinate of P is greater or less than the x -coordinate of M . [5]

[N17/II/4]

